**MicroAl-Dataset: Microstructure Images Across Aluminum Alloy Manufacturing Stages**

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Aluminum alloy microstructure images are **scarce and difficult to acquire** due to the high cost of material preparation, sophisticated imaging equipment, and the need for domain expertise in annotation. Unlike natural images that can be easily collected and labeled, each metallographic or electron microscope image requires careful sample preparation, precise instrument calibration, and expert interpretation. Moreover, annotations must be performed by materials scientists who can accurately identify phases, grains, and precipitates—a time-consuming and labor-intensive process.

To address these challenges, we present **MicroAl-Dataset**, a systematically curated collection of aluminum alloy microstructures covering **multiple manufacturing stages**. Rather than focusing on a single condition, our dataset spans the entire production workflow, enabling research on microstructure evolution across different processing steps. **Figure 1** illustrates the complete manufacturing process and the corresponding microstructure analysis pipeline:

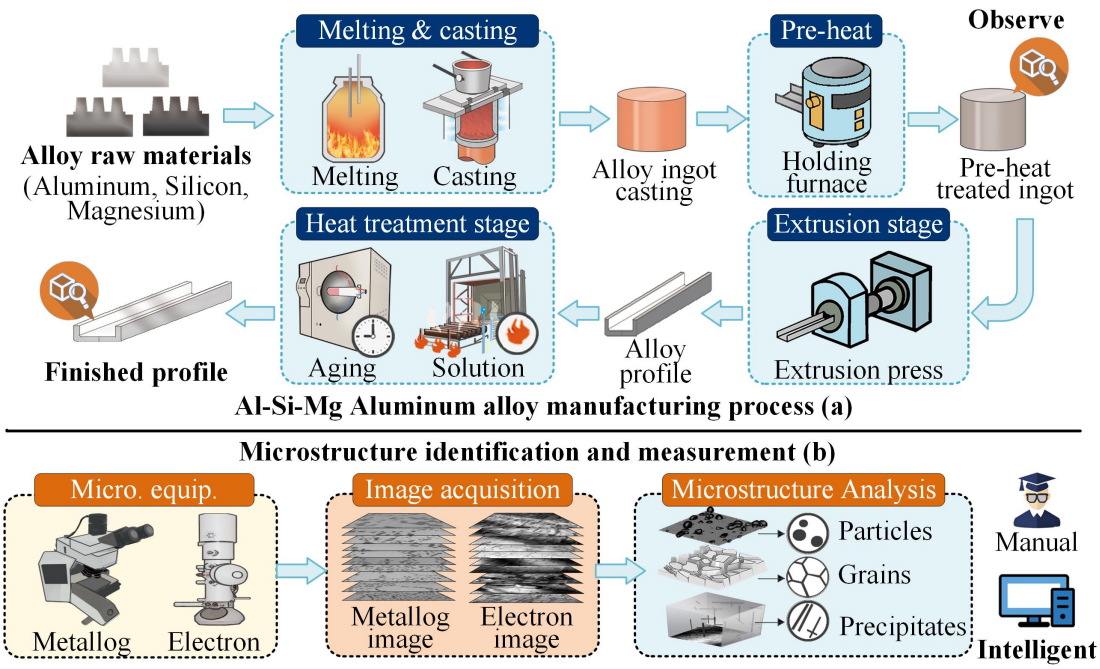


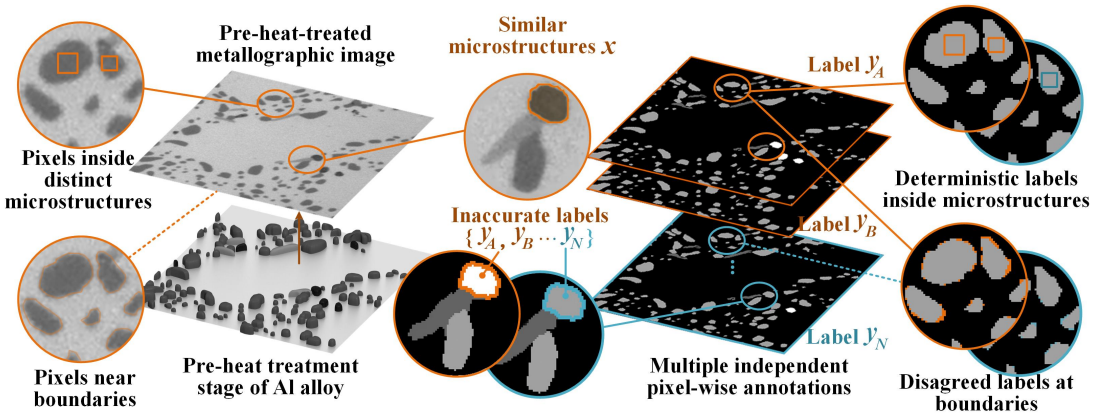

**Figure 1. Schematic diagram of aluminum alloy manufacturing process and microstructure analysis:** (a) Manufacturing workflow: casting, pre-heat treatment, hot extrusion, and heat treatment; (b) Analysis procedure: microscope observation, image acquisition, microstructure identification and measurement.

Based on this framework, we have constructed a total of **multiple microstructure datasets** covering different alloy systems, processing stages, and imaging modalities. Currently, **three of them have been publicly released**, as summarized in the table below. The remaining datasets are under active annotation or pending authorization and will be made available in future updates.

## 📂 Dataset Structure

| Level 1 | Level 2 | Level 3 | Description | Status |
|---|---|---|---|---|
| **Metallo** | **Pre-heat-treated** | pre-img<br>pre-gt | Metallographic images | ✅ Public **(1)** |
| | **Heat-treated** | image<br>mask<br>label | Metallographic images | ✅ Public **(2)** |
| | Nanshan | As-cast | As-cast aluminum alloys | ⏳ pending authorization |
| | | Film-coated | Film-coated aluminum grain | |
| | NEU-As-cast | 5xxx | 5xxx series alloys | ⏳ annotation in progress |
| | | 6xxx | 6xxx series alloys | |
| **Electron** | **TEM image** | TEM-img<br>annotation | Precipitate instance segmentation labels | ✅ Public **(3)** |
| | TEM dark-field image | - | TEM dark-field images of nano-scale precipitates | ⏳ annotation in progress |
| | HRTEM image | - | HRTEM images for precipitate phase identification | ⏳ annotation in progress |
| | Nanshan SEM image | - | SEM images of Nanshan aluminum alloy microstructures | ⏳ pending authorization |

### Dataset 1: Pre-heat-treated Metallographic Images – Second-phase Segmentation




This dataset focuses on **second-phase segmentation** in pre-heat-treated aluminum alloys.

- **Task**: Semantic segmentation of Si particles and Fe-bearing intermetallic phases.
- **Images**: Optical microscopy, 1024×1024 resolution (1 μm/pixel).
- **Annotations**: Pixel-wise labels supporting fully supervised and weakly supervised learning (incomplete/inaccurate).
- **Challenges**:
    - Similar contrast between Si particles and Fe-bearing phases.
    - Particle overlapping and occlusion.
    - Ambiguous phase boundaries due to etching effects.
    - Annotator disagreements at ambiguous regions.

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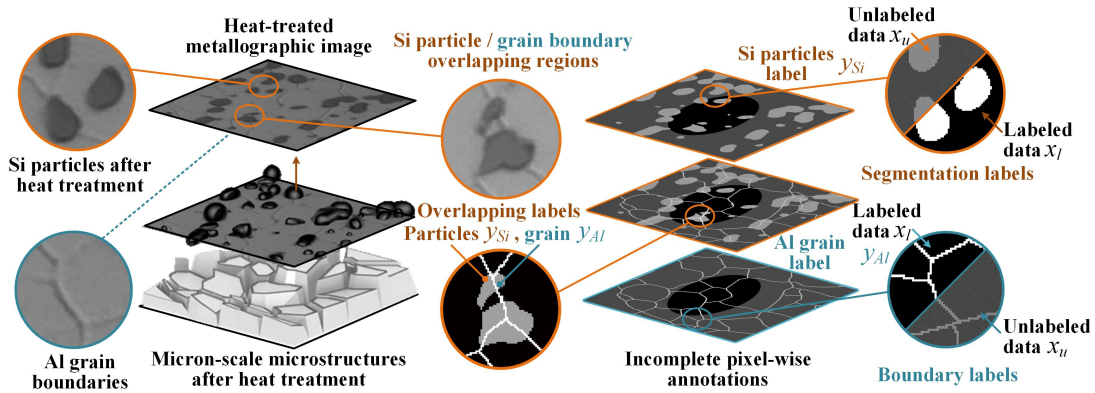
### Dataset 2: Heat-treated Metallographic Images – Grain Boundary Detection




This dataset addresses **grain boundary detection** in heat-treated microstructures.

- **Task**: Boundary detection for Al grains and semantic segmentation of Si particles.
- **Images**: Optical microscopy, 1024×1024 resolution (1 μm/pixel).
- **Annotations**: Boundary labels and segmentation masks (standard, inexact, incomplete).
- **Challenges**:
    - Grain boundaries partially occluded by Si particles.
    - Varying contrast across different grains.
    - Complex grain morphologies after recrystallization.
    - Incomplete boundaries due to weak etching or imaging artifacts.

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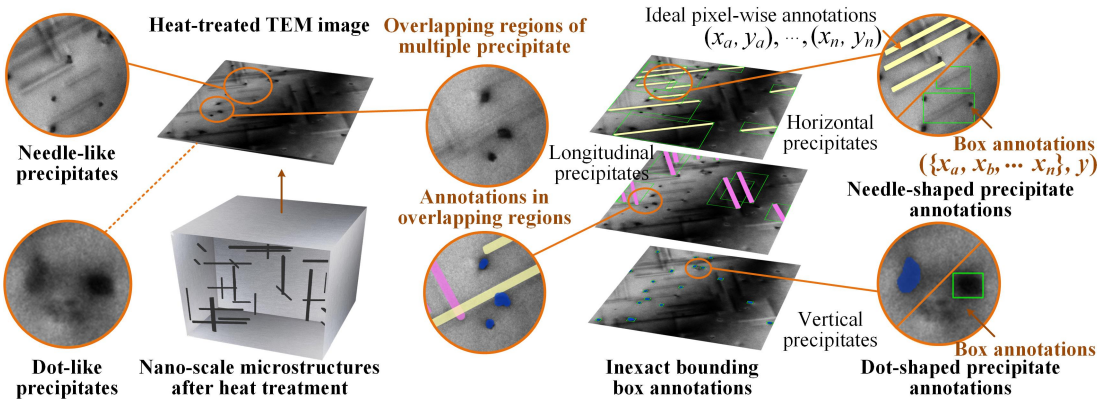
### Dataset 3: TEM Images – Instance Segmentation of Nano-scale Precipitates




This dataset focuses on **instance segmentation of nano-scale precipitates** under weak supervision.

- **Task**: Instance segmentation of precipitates ($Mg_2Si$, $Al_2Cu$, etc.).
- **Images**: Transmission electron microscopy (TEM), 2048×2048 resolution (1 nm/pixel).
- **Annotations**: Instance masks and bounding boxes (standard, inexact, inaccurate).
- **Challenges**:
    - High density of overlapping precipitates.
    - Low contrast in bright-field TEM images.
    - Ambiguous particle boundaries at the nano-scale.
    - Difficulty in distinguishing different precipitate phases.

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**Future Updates**

The datasets listed above are currently available. Additional images and annotations, including those under active annotation or pending institutional authorization, will be **continuously updated** upon completion. We are committed to expanding the dataset to cover more alloy systems, processing conditions, and annotation types to support the research community.